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Appliance and Equipment Standards Program
U.S. Department of Energy
Building Technologies Office
Mailstop EE-5B
1000 Independence Avenue SW
Washington, DC 20585-0121

**Re: Comments to Docket Number EERE-2021-BT-STD-0035
Proposed Energy Conservation Standards for Air Cleaners**

IQAir North America, Inc., together with Swiss affiliate IQAir AG (collectively "IQAir"), manufactures and distributes air purification systems for residential and commercial use. IQAir submits these comments to the 2023-04-11 Notice of Proposed Rule Making, Document ID EERE-2021-BT-STD-0035-0026 (the "Proposed Energy Conservation Standards for Air Cleaners", or the "Proposed Standards") to express concerns over the permanent negative effect the Proposed Standards will have on products that contribute to public health, and that are relied on and sought after by customers with special needs worldwide.

IQAir is a Swiss-based air quality technology company with a 60 year history, and U.S. headquarters near Los Angeles, CA for the past twenty years. IQAir employs approximately 75 workers in the U.S., and 500 worldwide. IQAir is committed to improving public health by making clean air available to all, and doing it in the most environmentally responsible ways possible.

IQAir's Successful Gas-Phase Filtration Systems

IQAir makes a variety of products for home, office and portable applications, and has developed a successful line of air purifiers for customers who need filtration of gases and odors. Some of these gases are Volatile Organic Compounds ("VOCs"). VOC's are airborne chemicals, often emitted by paints, cleaning supplies, and construction materials, which can cause severe short- and long-term health problems. Commonly

known VOCs include ammonia, benzene, and toluene. Other gases include formaldehyde, sulfur dioxide, and nitrogen dioxide.

Standard particulate and "HEPA" filtration systems do not reduce VOCs or other gas-phase contaminants. Instead, one of the most effective ways to filter gases and odors is with granular sorbent media such as activated charcoal, and chemisorbant pellets. The U.S. Environmental Protection Agency has stated:

"Gas-phase air-cleaning technologies include adsorbent media air filters such as activated carbon, chemisorbent media air filters, photocatalytic oxidation (PCO), plasma, and intentional ozone generators sold as air cleaners. Compared to the control of [particulate matter], gas-phase pollutant control is much more complex. Only adsorbent and chemisorbent media air filters have been shown to be effective gas-phase air cleaners for some gaseous pollutants without producing potentially harmful byproducts, although not all gaseous air pollutants are removed equally. Adsorbent media air filters have a finite capacity for adsorption and therefore must contain sufficient sorbent media for the application and must be replaced regularly."¹

IQAir has devoted decades of research and development to its granular filters, which include a proprietary blend of activated carbon and alumina, impregnated with potassium permanganate. These filters have allowed IQAir to create industry-leading gas-phase and VOC filtration systems.

At the top of IQAir's gas-phase filtration line is the GCX model. The GCX utilizes particulate pre-filtration, followed by cartridge-based granular filters containing over 20 pounds of media. These cartridge filters are available in five different configurations to address specific gas pollution challenges. The GCX's technology is similarly adapted to two other popular models: the GC MultiGas, and the HealthPro Plus. These models also offer excellent gas-phase performance, but at lower capacities, to address a broader segment of air purification customers.

Each of these three machines, the GCX, the GC MultiGas, and the HealthPro Plus, fall within the definition of "air cleaner" as proposed in the Energy Conservation Standards for Air Cleaners, and therefore would be governed by the new regulations. Unfortunately, under the proposed testing procedures, these machines will be measured no differently than standard "air cleaners" which offer no gas-phase filtration

¹ Residential Air Cleaners – A Technical Summary, 3rd ed., EPA 402-F-09-002, July 2018, page 6.

at all. The result is that due to the increased energy usage of gas-phase filtration, these machines will not meet the proposed "integrated energy factor", and therefore will no longer be able to be sold in the U.S.

Measuring Gas-Phase Performance

Air cleaners are commonly measured by their Clean Air Delivery Rate ("CADR"), which describes the initial cleaning performance of a filter. CADR is expressed with respect to specific types of pollutants. The Proposed Standards, for example, measure particulate matter, expressed as PM^{2.5}-CADR, and similarly provide a separate means for Pollen-CADR. It is also possible to determine CADR for gas-phase pollutants.

CADR, however, only measures initial air cleaning performance. It does not take into account the inherent degradation of performance over time. As air continuously passes through a filter, CADR decreases, because the filter becomes saturated. This is particularly relevant to gas-phase filtration, which relies on the *capacity* of granular media in order to maintain effective filtration. Without sufficient capacity, a granular filter might produce good initial gas-phase CADR, and then just hours later degrade to little or no filtration at all (while continuing to use just as much energy). Therefore, accurate measurement of gas-phase filtration must include capacity.

Currently, the most advanced standardized testing protocol for consumer gas-phase filtration is China's GB/T 18801-2022. This test measures not only initial CADR, but also the amount of pollutant removed from the air until CADR drops to 50% of the initial value (expressed as "CCM"). This methodology effectively measures the capacity of granular filters, enabling regulators and consumers to ensure that manufacturers do not game the system by achieving high CADR, or high energy efficiency, with unacceptable filter life.

The performance of the GCX on GB/T 18801-2022 is outstanding, registering a Formaldehyde-CADR rating of 300 m³/h, and a CCM of 27,500mg, exceeding the required minimum by over thirteen times. Likewise, the GC MultiGas and the HealthPro plus perform well above minimum requirements for their class. These features and performance are key considerations for regulators and users of these products.

The Value of Gas-Phase Filtration to Consumers

Clean air, and filtration of VOC's and other gaseous pollutants, are an important part of public health. IQAir's gas-phase machines have played a vital role for specific

segments of the population since their introduction, and they continue to see high demand from customers. One of the most relevant recent examples of this is the massive Aliso Canyon gas leak. In October 2015, in a suburb of Los Angeles, a natural gas storage facility operated by SoCal Gas Co. experienced a catastrophic failure, causing the largest natural gas blowout in U.S. history. Over 109,000 metric tons of natural gas were released into the air over a period of 111 days before it could be repaired. The natural gas also contained toxic VOCs such as benzene, and odorous gases such as mercaptan sulfur compounds. Thousands of residents were forced to evacuate their homes, and continue to suffer lasting health effects to this day.

SoCal Gas Co., working in concert with the California Air Resources Board, identified the IQAir GC MultiGas and HealthPro Plus as recommended devices for affected residents to mitigate the effects of the gas leak. These machines were specifically chosen for both their effectiveness at reducing VOCs and mercaptans, and their capacity. Previous testing conducted by the South Coast Air Quality Management District, in a 2012 study of air filtration in schools, showed that the HealthPro Plus is capable of a 52-73% removal performance for benzene in real-world classroom settings.² Moreover, the large capacity of the cartridges enabled these machines to absorb more contaminants over a longer period of time before needing filter replacements. IQAir provided over 10,000 devices to SoCal Gas for distribution to residents, and several thousand more machines directly to affected customers. The gas-phase filtration of these machines provided immediate, direct relief to those affected by a public health hazard.

Other real-world uses of IQAir gas-phase filtration abound. At the 9/11 museum in New York, which displays actual ground-zero artifacts, staff and visitors complained of headaches and other symptoms associated with off-gassing debris. Installation of a GC MultiGas and a HealthPro Plus had a drastic and immediate effect, completely eliminating the complaints. In Switzerland, the the HealthPro Plus and GC MultiGas is currently being used in school classrooms that were determined to have elevated levels of Polychlorinated Biphenyls (PCBs) as a result of degrading building materials. The air purifiers were proven to reduce these levels, and are being used as an intermediary solution because not all of the classrooms can be renovated at the same time.

Beyond customers who may use them for daily personal use due to their own unique circumstances, these machines often serve as a critical stopgap, reducing exposure and protecting users in situations where immediate remediation is not feasible. Whether it is the time that it took to cap a gas blowout, or the time that it takes to

² Polidori et. al., 2012, "Pilot study of high performance-air filtration for classroom applications."

renovate classrooms or retrofit a nearby factory, IQAir gas-phase filtration supports public health by offering unique, immediate benefits to users in need.

The Proposed Standards Would Effectively Ban Consumer Gas-Phase Filtration

The nature of gas-phase filtration inherently requires greater energy than simple particulate/HEPA filtration. Pushing air through the required pre-filtration, and then through varying types and amounts of granular media, requires electric motors of a certain power level. IQAir has carefully considered all available designs and technology when building and sourcing components, and has achieved what we believe is the best possible energy efficiency at given levels of gas-phase reduction, capacity, and price.

It is undeniable that the gas-phase filtration technology contained in IQAir's systems *performs a valuable, sought-after function*, and that standard particulate/HEPA systems *are physically incapable of performing this same function*. Likewise, there is no feasible combination of currently available components or technology that could allow these machines to meet the proposed standards, without going into a price range that is far out of reach of the people who depend on them.

Unfortunately, the proposed Energy Conservation Standards for Air Cleaners as currently drafted encompass a broad range of devices including gas-phase, but they measure only particulate performance. Gas-phase filters are unfairly measured, and their unique benefit is completely unrecognized. This application of the rules will eliminate an entire class of air purification products, and *deny some of our most vulnerable citizens from obtaining the gas-phase protection they want and need*. IQAir would be unable to provide machines such as the GC MultiGas to those affected by the next environmental disaster, and those with life-altering chemical sensitivities will be unable to purchase the machines for personal use.

Finally, these machines are critical parts of IQAir's product lineup, and the HealthPro Plus in particular is our single best-selling system. Prohibiting IQAir from offering these products, and from improving them as technology and circumstances allow, will be a devastating blow to IQAir's continued business operations in the United States. Moreover, a loss of the ability to sell these products in the United States would likely reduce overall sales volume overall to the point that it is no longer viable to remain in production even for sale abroad.

The Proposed Standards Should Recognize the Need For Gas-Phase Filtration

IQAir respectfully requests that the Department of Energy consider the importance of the entire class of gas-phase products that will be effectively banned by the broad stroke of the Proposed Standards. Including a means of measuring gas-phase performance and capacity, and a proportionate allowance in calculation of the integrated energy factor, would recognize the value these functions offer to users and the practical realities of implementing them, without reducing the effectiveness of the Proposed Standards on non-gas-phase air cleaners.

IQAir thanks the Department of Energy for its consideration, and would be happy to discuss the matter further.

Sincerely,



Glory Dolphin Hammes
CEO
IQAir North America, Inc.